

WAN Technologies — Guided Notes

Unit 1.8.3 | CompTIA Network+ N10-009 Obj. 1.8 | N10-008 1.8

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

Circuit switching establishes a _____ path for the duration of communication, guaranteeing _____ but wasting capacity when idle. Packet switching breaks data into _____ that share carrier infrastructure, which is more _____ but provides no dedicated bandwidth.

Question 2

A T1 leased line provides _____ Mbps. A T3 provides _____ Mbps. Leased lines offer _____ bandwidth and _____ performance. Their main disadvantage compared to modern alternatives is _____.

Question 3

MPLS routes traffic using _____ instead of IP addresses, creating virtual circuits. It supports _____ (two sites) and _____ (all sites can communicate directly) configurations. Its main advantage over broadband is _____ for latency, jitter, and packet loss.

Question 4

Metro Ethernet extends _____ connectivity beyond a single building using carrier _____. From the router's perspective it looks like _____. It is typically limited to _____ geographic areas.

Question 5

Match each broadband type to its key characteristic:

Cable → _____

DSL → _____

Fiber → _____

Question 6

An SD-WAN appliance at a branch office monitors _____, _____, and _____ on all WAN links in real time. Application-aware policies send _____ traffic over the highest-quality link and _____ traffic over cheaper broadband.

Question 7

The 'last mile' refers to _____. It is often the _____ in WAN performance. Installing fiber last mile to a building that doesn't have it can take _____, which affects WAN planning timelines.

Question 8 — Real World Application

REAL WORLD: Your organization has 20 branch offices each paying \$1,800/month for MPLS. A vendor proposes replacing each with \$150/month fiber broadband plus an SD-WAN appliance. What questions do you ask before approving the change?

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/1-8-3-wan-technologies/1-8-3-wan-technologies-answer-key.pdf
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